
2019-Ph-H1-Q18

Section: Particles and Waves

Topic: Interference

Question:

In Young's double-slit experiment, what causes bright fringes?

Answer:

Bright fringes occur when light from the two slits **interferes constructively** — i.e., **path difference = whole number of wavelengths**.

Correct Option: B (Path difference is a whole number of wavelengths)

Guidance for Students:

Dark = destructive = half wavelengths. Bright = constructive.

Revision Tips:

- Bright fringe: $\Delta d = n\lambda$
 - Dark fringe: $\Delta d = (n + \frac{1}{2})\lambda$
 - Know what path difference means!
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